MRI Super Resolution Challenge

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1 Introduction and background

Low field MRI systems provide an easily accessible and cost effective solution to bulky high cost MRI machines. However, it has its drawbacks; the resulting images can have a reduced resolution and weaker anatomical detail.This quality gap between the High field MRI machines and Low field ones can limit downstream analysis and reduces clinical usability in settings where high quality imaging is required.A practical approach is to computationally enhance LF scans to approximate HF like appearance while preserving underlying anatomy. This project formulates the LF to HF enhancement as a supervised image restoration task using paired LF and HF volumes. The pipeline first upsamples LF volumes to the HF grid to produce a baseline reconstruction, then learns a residual correction using a 2.5D convolutional network. The model consumes a stack of neighboring slices (this provides limited 3D context) and predicts a residual for the center slice, which is added back to the baseline. Training is aligned to evaluation-style normalization and perceptual metrics (SSIM / MS-SSIM), with Exponential Moving Average (EMA) weights and early stopping to improve stability since we have limited data.

2 Datasets

2.1 Dataset Description

The dataset consists of LF and their corresponding HF MRI volumes for the same subjects. LF volumes are lower resolution with fewer slices along the axial dimension, while HF volumes are higher resolution and contain substantially more slices. Each LF and HF pair: the LF volume acts as input and the HF volume acts as ground truth target.

2.2 Preprocessing

Each volume in the dataset is loaded from NIFTI format and normalized to [0,1] scaling using min-max normalization.Because LF and HF volumes differ in spatial resolution and slice count the LF volume is first resampled to the HF grid using trilinear interpolation, producing an upsampled baseline. Training samples are then generated by randomly cropping 2D patches from random spatial locations and slice indices. To incorporate limited through plane context, k adjacent slices centered at index z are stacked as channels to form a 2.5D input tensor of shape (k, patch, patch), while the target is the corresponding HF patch at the center slice z with shape (1, patch, patch). This patch based strategy increases the number of training samples and improves generalization when only a small number of volumes are available.

3 Methods

3.1 Problem Formulation

If we have a LF volume L the goal is to predict high field volume H. Since LF and HF volumes differ in resolution the LF volume is first resampled to HF grid via trilinear interpolation to obtain a baseline

$$H_0 = Upsample(L)$$

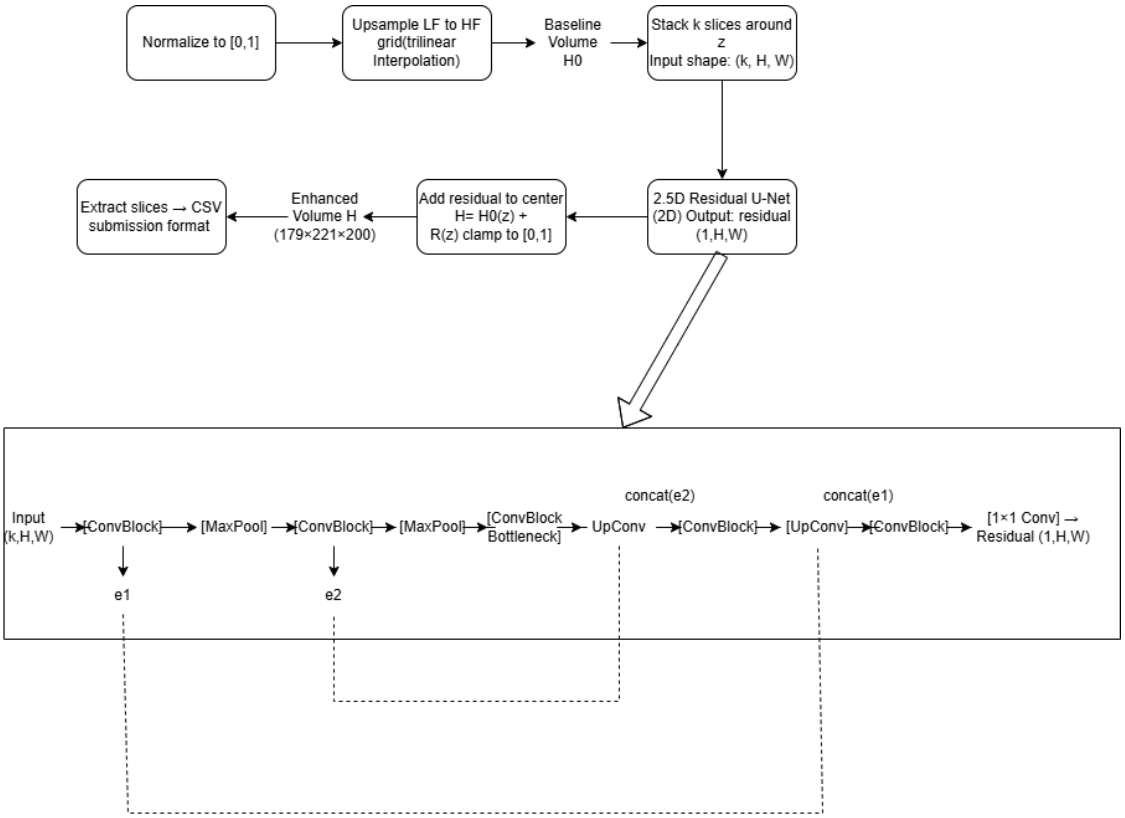
The model is trained to predict a residual correction R for the center slice of a k slice context window and the final out will be

$$H = H_0 + R$$

Residual learning helps the network focus on missing high frequency details than the entire image intensity distribution.

3.2 Model Architecture

A 2.5D resiual U-NET is used. The input to the network is a stack of k neighboring slices from the upsampled baseline.The U-NET comprises of an encoder decoder design with skip connections:convolution blocks extract multi-scale features, pooling downsamples to a bottleneck representation, and transpose convolutions upsample back to the original resolution. Skip connections concatenate encoder feature maps into the decoder to preserve spatial detail. The final layer is a 1×1 convolution producing a single-channel residual map for the center slice. The output residual is added to the center baseline slice, and the result is clamped to [0, 1] for numerical stability and consistency with normalization.



Architecture Diagram:

3.3 Loss Functions and Metric Alignment

To reflect evaluator behavior, normalization is applied per-slice (or per-sample) prior to computing perceptual losses. Two training configurations were explored. In the SSIM+MSE configuration, the loss is a weighted sum of 1 minus SSIM (computed using a local Gaussian window) and mean squared error (MSE) on normalized slices. In the MS-SSIM+L1 configuration, multi-scale structural similarity (MS-SSIM) is used to better capture perceptual quality across multiple spatial scales, combined with L1 loss for intensity fidelity. In both cases, metric-aligned normalization reduces sensitivity to global intensity scaling differences and encourages optimization of structural similarity.

3.4 Training

Training is performed using Adam optimization with a fixed learning rate and random patch sampling. An Exponential Moving Average (EMA) copy of the model weights is maintained and used for validation, improving stability under small-data training. Early stopping is applied based on a lightweight validation proxy computed on a small set of slices per validation volume. This avoids overfitting and reduces unnecessary training iterations.

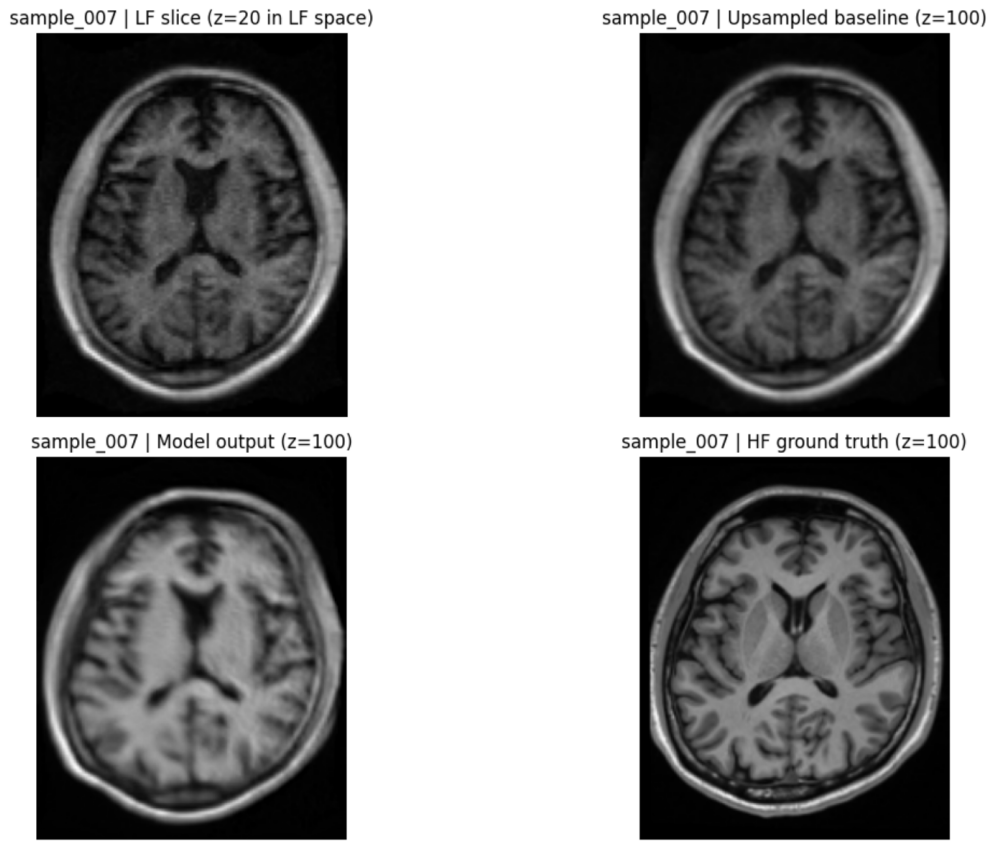
3.5 Inference

At inference time, the LF test volume is normalized and resampled to the HF grid to produce the baseline volume. For each HF slice index z, the method forms a k-slice input stack from neighboring baseline slices, runs the network in batches of slices for efficiency, and reconstructs the full enhanced volume by placing each predicted center slice into the output volume. The final prediction volume is clipped to [0, 1] and formatted into the required submission CSV representation by extracting slices in the evaluator's expected order.

4 Results

4.1 Training Details

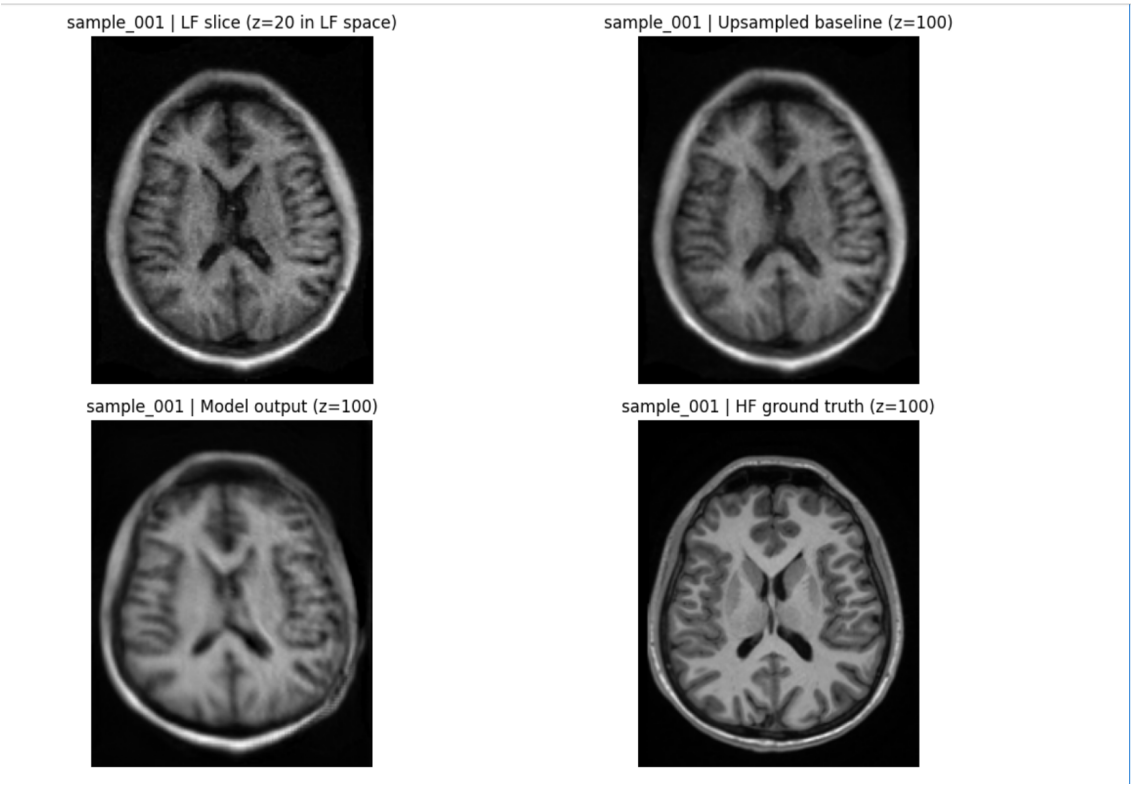
Experiments were conducted with a small validation split (e.g., two volumes) due to limited data availability. Patches of size 128×128 were sampled randomly. Typical hyperparameters include batch size 4, learning rate in the range of 1e-4 to 2e-4, and k-slice context windows such as k=5,k=7,k=9. Validation was performed periodically using EMA weights, with early stopping based on a proxy structural similarity score computed on evenly spaced slices.



The Output of the first half i.e. SSIM centered:

4.2 Results

Quantitative evaluation compares the upsampled baseline to the model enhanced output using proxy metrics aligned with evaluation normalization. Across validation slices, the model improves perceptual similarity over the baseline, reflected in higher SSIM/MS-SSIM proxy scores and improved PSNR on normalized slices. The use of residual learning and 2.5D context contributes to better preservation of anatomical edges and reduction of blurring artifacts introduced by interpolation.Qualitative inspection shows that the upsampled baseline often appears overly smooth, with reduced edge sharpness and loss of



The Output of the second half i.e. MS-SSIM centered:

fine structural detail. The proposed enhancement model improves local contrast and edge definition and reduces visible interpolation artifacts.

5 LLM Prompts

The LLM was used to support implementation and debugging. Usage included: clarifying the behavior of SSIM/MS-SSIM, verifying tensor shapes for 2.5D slicing and batching, handling edge cases in slice indexing, and improving code organization. Prompts included "Explain why MS-SSIM has a minimum image-size constraint and how to select an appropriate window size," and "Describe how to map LF slice indices to corresponding HF indices for visualization."

6 Conclusions

This project demonstrates that a 2.5D residual U-Net can improve LF→HF enhancement over a simple interpolation baseline, particularly in terms of perceived sharpness and structural similarity. Metric-aligned normalization and perceptual losses help focus optimization on anatomical structure rather than purely pixel-wise differences. EMA and early stopping improve stability and reduce overfitting risk when training data is limited.Limitations include the small number of training volumes, potential slice wise inconsistency due to 2D prediction, and reliance on interpolation to align LF and HF grids. Future work may explore full 3D architectures, improved spatial alignment, stronger regularization, and larger training datasets to further improve generalization and volumetric consistency.

7 Author contributions

I am the sole contributor of this project.